6.8 Graphs of Polar Equations

Name _____ Period ____ Date _____

Pre-Calculus 6.8 · Student sheet

I can…

- test a polar equation for symmetry.
- find the zeros and maximum |r| of a polar equation.
- recognize and sketch limaçons and rose curves.

Vocabulary · fill in the missing word

Limaçon — the _____ of $r = a \pm b\cos\theta$ or $r = a \pm b\sin\theta$

Cardioid — a _____çon with $a = b$ (heart-shaped)

Rose curve — the _____ of $r = a\cos n\theta$ or $r = a\sin n\theta$, $n \geq 2$

Lemniscate — the _____ of $r^2 = a^2\sin 2\theta$ or $r^2 = a^2\cos 2\theta$

Part 1 · A polar gallery

Graph each one in Desmos and sketch it.

Equation	Sketch / description
$r = 2 + 2\cos\theta$	
$r = 1 + 2\cos\theta$	
$r = 3 + \cos\theta$	
$r = 2\cos 3\theta$	
$r = 2\cos 2\theta$	

Part 2 · Notes and examples (fill in each note, then work the example)

Symmetry tests: replace (r, θ) with $(r, -\theta)$ for the polar axis, $(r, \pi - \theta)$ for the line $\theta = \pi/2$, and $(r, \theta + \pi)$ or $(-r, \theta)$ for the _____.

Example 1

Test $r = 3 + 2\cos\theta$ for symmetry.

Zeros and maximum |r|: find the θ-values where $r = 0$ and where $|r|$ is _____.

Example 2

Find the zeros and the maximum $|r|$ of $r = 1 - 2\cos\theta$ $(0 \leq \theta < 2\pi)$.

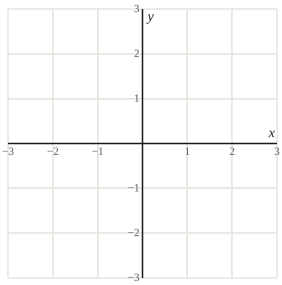
Rose curves $r = a\cos n\theta$ or $a\sin n\theta$: n petals if n is odd, _____ petals if n is even; each petal is $|a|$ long.

Example 3

Describe each rose curve.

a. $r = 2 \cos 3\theta$

b. $r = 3 \sin 2\theta$



Limaçons $r = a \pm b \cos \theta$: $a/b < 1$ inner loop; $a/b = 1$ _____; $1 < a/b < 2$ dimpled; $a/b \geq 2$ convex.

Example 4

Classify each limaçon.

a. $r = 2 + 2 \sin \theta$

b. $r = 1 + 2 \cos \theta$

c. $r = 3 + 2 \cos \theta$

d. $r = 5 + 2 \cos \theta$

Example 5

Identify the graph of $r^2 = 4 \sin 2\theta$.

Part 3 · Partner practice

Partner 1

Describe $r = 4 \cos 5\theta$.

Partner 2

Classify $r = 3 - 3 \sin \theta$, give its symmetry, and find its maximum $|r|$.